

Homework 1

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Exercise 1

Part (a)

Suppose $0 \mid 10$. There then exists some $c \in \mathbb{Z}$ such that $0 \cdot c = 10$. However, $0 \cdot c = 0$ for all c . This is a contradiction; therefore $0 \nmid 10$.

Part (b)

Since $15 \cdot c = 2025$ for $c = 135 \in \mathbb{Z}$, $15 \mid 2025$.

Part (c)

Note that -285714 can be uniquely expressed as $17 \cdot -16806 + 5$. Since $5 \neq 0$, $17 \nmid -285714$.

Exercise 2

Part (a)

Consider the case where $a = 2, b = 4, c = 3, d = 9$. Note that $a \mid b$ and $c \mid d$. However, $5 \nmid 13$, so $a + c \nmid b + d$. Therefore it is not necessarily the case that for $a, b, c, d \in \mathbb{Z}$ such that $a \mid b$ and $c \mid d$, that $a + c \mid b + d$.

Part (b)

Consider the case where $a = 9, b = 3, c = 6$. Note that $a \mid bc$. However, $9 \nmid 3$ and $9 \nmid 6$, so $a \nmid b$ and $a \nmid c$. Therefore it is not necessarily the case that for $a, b, c, d \in \mathbb{Z}$ such that $a \mid bc$, that $a \mid b$ or $a \mid c$.

Part (c)

Consider the case where $a = 3, b = 4, c = 6$. Note that $a \nmid b$ and $b \nmid c$. However, $3 \mid 6$, so $a \mid c$. Therefore it is not necessarily the case that for $a, b, c, d \in \mathbb{Z}$ such that $a \nmid b$ and $b \nmid c$, that $a \nmid c$.

Exercise 3

Let $n \in \mathbb{Z}$. Then $n = 5 \cdot q + r$ where $0 \leq r < 5$ for unique $q, r \in \mathbb{Z}$. Then:

$$\begin{aligned} n^5 - n &= (5q + r)^5 - (5q + r) \\ &= (5q)^5 + 5(5q)^4 r + 10(5q)^3 r^2 + 10(5q)^2 r^3 + (5q)r^4 + r^5 - 5q - r \\ &= 3125q^5 + 3125q^4 r + 1250q^3 r^2 + 250q^2 r^3 + 5qr^4 + r^5 - 5q - r \\ &= 5(625q^5 + 625q^4 r + 250q^3 r^2 + 50q^2 r^3 + qr^4 - q) + r^5 - r \end{aligned}$$

Suppose $r = 0$; then $r^5 - r = 0$, and $5 \mid 0$. Suppose $r = 1$; then $r^5 - r = 0$, and $5 \mid 0$. Suppose $r = 2$; then $r^5 - r = 30$, and $5 \mid 30$. Suppose $r = 3$; then $r^5 - r = 240$, and $5 \mid 240$. Suppose $r = 4$; then $r^5 - r = 1020$, and $5 \mid 1020$. In all cases, $r^5 - r = 5 \cdot c$ for some $c \in \mathbb{Z}$. Then:

$$\begin{aligned} n^5 - n &= 5(625q^5 + 625q^4 r + 250q^3 r^2 + 50q^2 r^3 + qr^4 - q) + 5c \\ &= 5(625q^5 + 625q^4 r + 250q^3 r^2 + 50q^2 r^3 + qr^4 - q + c) \end{aligned}$$

Consequently, $5 \mid n^5 - n$.

Exercise 4

Part (a)

The primes less than $\sqrt{201}$ are 2, 3, 5, 7, 11, 13.

p	$pq + r = 201$
2	$2 \cdot 100 + 1$
3	$3 \cdot 67 + 0$
5	$5 \cdot 40 + 1$
7	$7 \cdot 28 + 5$
11	$11 \cdot 18 + 3$
13	$13 \cdot 15 + 6$

Since $3 \mid 201$, 201 is not prime.

Part (b)

The primes less than $\sqrt{211}$ are 2, 3, 5, 7, 11, 13.

p	$pq + r = 201$
2	$2 \cdot 105 + 1$
3	$3 \cdot 70 + 1$
5	$5 \cdot 42 + 1$
7	$7 \cdot 30 + 1$
11	$11 \cdot 19 + 2$
13	$13 \cdot 16 + 3$

Since $p \nmid 211$ for all prime $p < \sqrt{211}$, 211 is prime.

Part (c)

The primes less than $\sqrt{213}$ are 2, 3, 5, 7, 11, 13.

p	$pq + r = 201$
2	$2 \cdot 106 + 1$
3	$3 \cdot 71 + 0$
5	$5 \cdot 42 + 3$
7	$7 \cdot 30 + 3$
11	$11 \cdot 19 + 4$
13	$13 \cdot 16 + 5$

Since $3 \mid 213$, 213 is not prime.

Part (d)

The primes less than $\sqrt{221}$ are 2, 3, 5, 7, 11, 13.

p	$pq + r = 201$
2	$2 \cdot 110 + 1$
3	$3 \cdot 73 + 2$
5	$5 \cdot 44 + 1$
7	$7 \cdot 31 + 4$
11	$11 \cdot 20 + 1$
13	$13 \cdot 17$

Since $13 \mid 221$, 221 is not prime.

Exercise 5

Suppose $p = 3n + 1$ for some $n \in \mathbb{N}$. Suppose n is odd, and so $n = 2k + 1$ for some $k \in \mathbb{N}$. Then:

$$\begin{aligned} p &= 3n + 1 \\ &= 3(2k + 1) + 1 \\ &= 6k + 3 + 1 = 6k + 4 \\ &= 2(3k + 2) \end{aligned}$$

Consequently, $2 \mid p$ and so p is even. However, the only even prime is 2, which is impossible since that means that $n = \frac{1}{3}$. This is a contradiction; therefore n is even and so $n = 2k$ for some $k \in \mathbb{N}$. Then:

$$\begin{aligned} p &= 3n + 1 \\ &= 3(2k) + 1 \\ &= 6k + 1 \end{aligned}$$

Consequently, $p = 6n + 1$ for some $n \in \mathbb{N}$.

Exercise 6

Let $a, n \in \mathbb{N}$ such that $a, n > 1$ and $a^n - 1$ is prime. We first show that for $k, s \in \mathbb{N}$, $a^{ks} - 1 \mid a^k - 1$ for all s .

Clearly the base case where $s = 1$ holds, as $a^k - 1 \mid a^k - 1$. Suppose $a^k - 1 \mid a^{ks} - 1$ for some $s \in \mathbb{N}$ and so $a^{ks} - 1 = (a^k - 1) \cdot c$ for some $c \in \mathbb{Z}$. Then:

$$\begin{aligned} a^{k(s+1)} - 1 &= a^{k(s+1)} - a^{ks} + a^{ks} - 1 \\ &= a^{ks}(a^k - 1) + (a^k - 1) \cdot c \\ &= (a^k - 1)(a^{ks} + c) \end{aligned}$$

Therefore $a^k - 1 \mid a^{k(s+1)} - 1$ and by induction, $a^k - 1 \mid a^{ks} - 1$ for all s .

If $a^n - 1$ is prime, since the only divisors of n are 1 and itself, it must be the case that $a - 1 \mid a^n - 1$. Note that if $a \neq 2$, some integer greater than 1 ($a - 1$) divides $a^n - 1$ and so $a^n - 1$ cannot be prime. Therefore for $a^n - 1$ to be prime, $a = 2$.

Now suppose n is composite; then $n = ks$ for some $k, s \in \mathbb{Z}$ such that $1 < k, s < n$. Then $a^k - 1 \mid a^n - 1$. Since $a = 2$ and $k > 1$, $a^k - 1 > 1$. Then some integer greater than 1 ($a - 1$) divides $a^n - 1$ and so $a^n - 1$ cannot be prime. This is a contradiction; therefore n is prime.